

ARTICLE

Study longer or study effectively? Better study strategies can compensate for less study time and predict goal achievement and lower negative affect

Maria Theobald 

Trier University, Trier, Germany

Correspondence

Maria Theobald, Trier University,
Universitätsring 15, 54296 Trier, Germany.
Email: theobald@uni-trier.de

Funding information

Center for School, Education, and Higher
Education Research, Johannes Gutenberg
University Mainz

Abstract

Background and Aims: The hypothesis that study strategies can compensate for less study time in predicting learning outcomes has often been proposed but rarely tested empirically.

Methods: In the present study, 231 university students reported their daily perceived time spent on self-study, study strategies (planning, monitoring, concentration and procrastination) and goal achievement over a 30 days period.

Results and Conclusion: Results showed that both more overall perceived study time and better study strategies (better planning, monitoring, and concentration, less procrastination) predicted higher goal achievement at the end of the day. In addition, perceived study time and study strategies interactively predicted goal achievement. When students reported better planning, monitoring and concentration as well as lower procrastination, less time was needed to achieve a high goal level compared to days on which they studied less strategically. In other words, when students studied less strategically, they had to invest more time to reach a higher goal level. In addition, perceived study time and study strategies were related to students' negative affect. Negative affect was particularly high when students studied for many hours with low concentration, and it was particularly low when students studied for only a few hours and procrastinated less. Taken together, the results suggest a compensatory effect of study time and study strategies on daily goal achievement and affect, highlighting the need to teach students effective study strategies.

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial](https://creativecommons.org/licenses/by-nc/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited and is not used for commercial purposes.

© 2024 The Author(s). *British Journal of Educational Psychology* published by John Wiley & Sons Ltd on behalf of British Psychological Society.

KEYWORDS

affect, ambulatory assessment, goal achievement, higher education, self-regulated learning, study strategies, study time

INTRODUCTION

Using study time efficiently should be better than studying for many hours for at least two reasons. First, when time is used efficiently by using good study strategies, less time is needed to achieve the goal. This efficient use of time is also referred to as good time management (Wolters & Brady, 2020), which in turn is associated with better academic achievement (Broadbent & Poon, 2015; Richardson et al., 2012). In addition to the positive effects on academic achievement, better time management has also been associated with more positive affective outcomes and lower levels of stress (Häfner et al., 2014; Macan et al., 1990). On the other hand, a high workload in terms of many study hours can increase negative affect and decrease student well-being (Galloway et al., 2013; Theobald & Bellhäuser, 2021). Thus, effective use of study time may not only lead to improved academic achievement but also reduce negative affect and stress. Therefore, the present study seeks to answer the question of how study time and study strategies interact in predicting daily goal achievement and affect.

Study strategies can be classified using models of self-regulated learning (for a review see Panadero, 2017). The present study focused on two types of self-regulated learning strategies, metacognitive strategies and resource management strategies (Boekaerts, 1999). First, metacognitive strategies refer to second-order cognitions that serve to monitor and control strategy use (see Veenman et al., 2006). Planning and self-monitoring include the most important metacognitive processes that take place before learning (i.e. planning one's learning tasks) as well as during and after learning (i.e. self-monitoring one's understanding and progress towards goals). Second, resource management strategies refer to strategies used to regulate internal resources (e.g. attention, effort or motivation) and external resources (e.g. learning environment or time) (Pintrich, 1999). Low procrastination and high concentration involve the ability to effectively manage one's internal and external resources to initiate and maintain task completion. For example, a student who procrastinates has failed to self-motivate and manage time and effort to initiate a task. A student who fails to concentrate has failed to maintain the motivation and effort to continue working on the task. Taken together, these metacognitive and resource management strategies are important for successfully initiating (planning, not procrastinating) and sustaining (monitoring, staying focused) self-regulated learning processes and are thus the focus of the present investigation.

How should study strategies and study time interact in self-regulated learning models? The component model of self-regulated learning (Schmitz & Wiese, 2006) emphasizes that study time should lead to good learning outcomes only if it is used effectively, for example, by using the aforementioned metacognitive strategies to plan and monitor one's own learning and resource management strategies to stay focused during learning and avoid procrastination (Boekaerts, 1999; Pintrich, 1999). According to this theoretical assumption, learning quantity (in terms of study time) and learning quality (in terms of study strategies) should interactively predict learners' goal achievement after learning. However, the exact nature of this interaction for specific study strategies is not specified. Furthermore, it remains unclear whether learning quantity and learning quality also predict learners' affective responses after learning. For example, learners may feel happy when they achieve their goal or disappointed when they study for many hours without achieving their goal. Taken together, according to models of self-regulated learning, the quantity and quality of study time—and, more specifically, their interaction—should predict whether learners achieve their goals in a given study session. What remains unclear, however, is exactly how this interaction occurs.

Previous empirical research has documented associations between study time, study strategies and academic achievement. With respect to study time, prior research suggests an average positive but small correlation between study time and academic achievement (Doumen et al., 2014; Landrum et al., 2006;

Plant et al., 2005). A meta-analysis of studies found an average correlation between .15 and .19 for the relation between study time and overall grade point average; the relation between study time and individual course grades was close to zero ($r = .01$) (Credé & Kuncel, 2008). With respect to study strategies, meta-analyses have found that better planning, self-monitoring and concentration, as well as less procrastination, are associated with better academic achievement (Broadbent & Poon, 2015; Dent & Koenka, 2016; Richardson et al., 2012; Steel, 2007). Thus, previous research suggests that, on average, studying longer and studying strategically is associated with better academic achievement.

While these meta-analyses examined learning quantity and quality separately, several studies tested study time alongside study strategies as predictors of academic achievement (e.g., Doumen et al., 2014; Plant et al., 2005; Theobald et al., 2018). For example, Plant et al. (2005) found that study time did not predict grade point average when study environment was controlled for. It was hypothesized that studying alone in the library would be a better study environment than studying with others or at home, as the latter could be distracting. These results suggest that students should choose a study environment that allows them to concentrate rather than spend more hours studying. Another study assessed study strategies more explicitly by looking at how well students distributed their learning over the semester and whether they used self-tests to monitor their understanding (Theobald et al., 2018). Supporting previous finding, the study found that when distributed learning and the use of self-tests were controlled for, time spent studying no longer predicted students' exam grades. Doumen et al. (2014) even tested the proposed interactive effect of study time and study strategies. They found that those students who completed fewer exam preparation exercises needed more study time to achieve the same course grade as those who completed many exercises. Hence, higher use of time compensated for lower use of study strategies, in this case completing practice problems. Thus, these results suggest that the quality of learning, in terms of strategy use, is more important than the amount of time spent studying in predicting academic achievement.

However, these previous studies did not test whether and how study time and study strategies were related to students' daily goal achievement. That is, these studies tested how total study time or average study strategies correlated with an aggregate measure of students' academic achievement, namely, end-of-semester course grades or grade point average. However, students' study time and study strategies, and their goal achievement, can vary from one study session to the next (Liborius et al., 2019; Theobald et al., 2021). Therefore, a more fine-grained assessment of students' variations in study time and study strategies is needed to explain day-to-day variations in goal achievement.

A more fine-grained assessment of variations in study time, study strategies and goal achievement would also allow testing of within-person relations. Self-regulated learning models describe the role of learning quantity and quality on a much finer time scale, within a study session (Schmitz & Wiese, 2006). Moreover, self-regulated learning is a process that takes place within individuals, but previous research has mainly focused on relations at the between-subjects level. However, the functional relations between variables may differ between the two levels of analysis (Molenaar, 2004). Therefore, previous findings from studies examining interindividual differences in study time, study, strategies and goal achievement at the between-person level may not apply to intraindividual differences at the within-person level. Given that the goal of the present study was to examine the dynamic, daily interplay between study time, study strategies, goal achievement and affect, analyses focused on the within-person level.

There are few studies that have examined the association between study time, study strategies and goal achievement on a daily basis. A recent study that synthesized data from five independent studies using repeated assessments found no correlation between study time and daily goal achievement (Theobald et al., 2023). In contrast, procrastination, which can be viewed as a more qualitative measure of how study time is used, was related to daily goal achievement. Another study examined how daily time investment predicted daily satisfaction with the study day, which can be viewed as a correlate of goal achievement (Liborius et al., 2019). This study found a small, albeit significant, positive correlation between study time and daily study satisfaction. However, this study also found that planning strategies, procrastination and effort investment were much more strongly related to daily study satisfaction. Taken together, these findings suggest that how students study, in terms

of their strategies, may be more important than how many hours they study. Importantly, however, none of these studies tested the hypothesized interactive effect of study time and study strategy use on daily goal achievement. Thus, the central question of this paper remains unanswered: Can better strategies offset shorter study time?

If effective use of study time promotes goal achievement in less time, it may also have beneficial effects on students' overall study-related negative affect. As noted above, high levels of study time have been associated with higher levels of negative affect (Galloway et al., 2013; Theobald & Bellhäuser, 2021). For example, one study found that study load and negative affect increased in parallel over the course of a semester (Theobald & Bellhäuser, 2021). Importantly, negative affect was particularly high when students studied longer than they had planned on a given day. These findings suggest that the negative relation between a high number of study hours and negative affect may be exacerbated if time is used inefficiently—a hypothesis that has not been empirically tested, however. Similar conclusions can be drawn from studies focusing on procrastination, which involves the voluntary postponement of intended actions (Steel, 2007). For example, students who procrastinate may spend a large number of hours at their desks without actually working on their intended study task, indicating inefficient use of study time. High levels of procrastination have consistently been associated with negative affect and stress (Sirois, 2023; Tice & Baumeister, 1997). Taken together, these previous studies suggest that many study hours increase negative affect, especially when time is used ineffectively.

THE PRESENT STUDY

The present study addresses the question of how study time and study strategies interact in predicting daily goal achievement and affect at the within-person level. It is hypothesized that on days when students use better study strategies (more planning and monitoring, better concentration and less procrastination compared to their own average), they will report higher daily goal achievement in less study time than usual compared to days when students study less strategically (H1). A similar interaction between study strategies and study time is expected for negative affect as an outcome. It is hypothesized that students will report less negative affect on days when they study less than usual but also use better study strategies than on days when they study more than usual but less strategically (H2). Based on self-regulated learning models and previous research, it was not possible to formulate differentiated hypotheses for specific study strategies *a priori*. To account for the dynamic, within-person interplay between study time, study strategies, goal achievement and affect, this study uses daily assessments of the focal variables. This assessment procedure, which takes place in students' natural study environment, further reduces potential biases that could occur when students are asked to retrospectively recall how and how much they studied during the semester (Klug et al., 2011). Taken together, this study advances our understanding of the interactive within-person interplay between study time, study strategies in predicting daily goal achievement and study-related affect.

METHODS

Participants

Initially, 257 participants were recruited through flyers and advertisements on the campus of a large university in Germany. Prior to enrolment, students provided informed consent. Students were informed of the study procedure and that the purpose of the study was to examine how they prepare for their exams. Twenty-six participants were excluded because they withdrew before the start of the ambulatory assessment period or did not provide a single complete data point. The final sample consisted

of 231 students who were on average 22 years old ($M=21.99$, $SD=2.29$, [18; 31]; 74% female), in their fourth semester ($M=3.66$, $SD=2.20$, [1; 10]) and came from different fields of study.

Procedure and design

The study included a pre-questionnaire, a daily ambulatory assessment period over the course of 30 days, and a post-questionnaire (see Figure 1). The study was administered using SoSci Survey (Leiner, 2019). The pre-questionnaire included demographic information about the participants, and several questionnaires about their self-regulated learning, motivation and past achievements. The daily ambulatory assessment period included daily morning and evening questionnaires that asked participants about their daily self-regulated learning (see Section 3.3). Students received daily email invitations to complete the morning and evening questionnaires. In addition, the ambulatory assessment period included an experimental manipulation of feedback provision (for details, see Theobald & Bellhäuser, 2022). The post-questionnaire included the same questionnaires as the pre-questionnaire and was administered after the daily ambulatory assessment period was completed. In this paper, only selected measures from the daily ambulatory assessment period are reported but a full list of all measures assessed is publicly available (<https://osf.io/r9n4y/>).

Measures

All variables reported below were assessed during the 30 days ambulatory assessment period. In the morning questionnaire, students were asked to describe their academic goals in an open-ended text box. Then, students reported their planning strategies (3 items, e.g. ‘Today, I have a concrete plan for how far I want to get with my learning’, within-subject $\omega = .65$, ICC = .41). In the evening questionnaire, students reported their self-monitoring strategies (3 items, e.g. ‘Today, I paid attention to how far I am from my learning goal’, within-subject $\omega = .56$, ICC = .35), procrastination (3 items, e.g. ‘Today I put off finishing a task’, within-subject $\omega = .70$, ICC = .22) and concentration (3 items, e.g. ‘Today I was unfocused while studying’, within-subject $\omega = .84$, ICC = .24). Students also reported their goal achievement (‘Today, I achieved my goals.’; range 0%–100%, ICC = .23) and their perceived study time for self-study in hours in an open-ended text box (ICC = .32). Daily questionnaire items were adapted from established trait questionnaires, but slightly rephrased to refer to daily states (e.g. Glöckner-Rist, et al., 2014; Wild & Schiefele, 1994; for full item lists and comparisons, see <https://osf.io/r9n4y/>). The daily measures were substantially correlated with corresponding trait measures assessed once at pretest (see Table S1), supporting the construct validity of the daily measures. The questionnaire items were rated on a 6-point Likert scale ranging from ‘not true’ to ‘true’. In addition, students’ daily affect was assessed in the evening questionnaire using a short version of the positive and negative affect (Schallberger, 2005; 10 items, within-subject $\omega = .83$, ICC = .35,

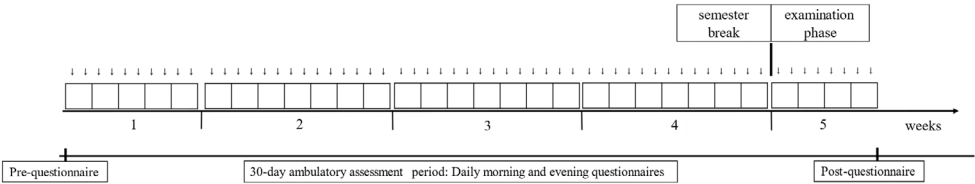


FIGURE 1 Overview of the survey period. Students completed a pre-questionnaire, a 30 days ambulatory assessment period and a post-questionnaire. During the daily ambulatory assessment period, students completed one morning and one evening questionnaire over the course of 30 days, as indicated by two arrows (↓) per box.

6-point scale). Higher scores on the affect scale generally indicate more negative affect (e.g. dissatisfied, stressed and tired).

Data analysis

Data were analysed in R with a significance level of $\alpha = .05$. The data and the data analysis script are publicly available (<https://osf.io/r9n4y/>). Hierarchical linear modelling was used to account for the nested data structure (measurement points clustered within participants). Experimental condition was the only between-subjects variable included as a control variable in all analyses. Otherwise, all variables were within-person variables that were person-mean centred to refer to within-person relations. The same procedures were used once with goal achievement and once with affect as the outcome variable. Predictor variables were entered in three steps: First, only perceived study time and experimental condition (as control variable) were entered. Second, the four study strategy variables were entered. Third, four interaction terms for the study strategies with perceived study time were entered. For data analysis, $k = 131$ data points were excluded because students self-reported in a control question that they did not answer the questionnaire conscientiously that day. Another $k = 229$ datapoints were excluded because students reported that they did not study (i.e. study time equalled '0'; 5% of data). In addition, one outlier data point was excluded because the student reported a study time of 20 h a day, which may not be realistic and was significantly different from the rest of the distribution (>5 SD above average reported study time). The final number of data points included in the analyses was $k = 3411$.

RESULTS

Preliminary results

The present investigation used data from a study testing the effects of a feedback intervention (for details see Theobald & Bellhäuser, 2022). Students in a metacognitive feedback condition ($n = 61$) received daily written feedback on their use of metacognitive strategies. Students in the motivational feedback condition ($n = 61$) received daily written feedback on their motivation. Students in the metacognitive and motivational feedback condition ($n = 62$) received daily written feedback on both their metacognitive strategy use and their motivation. Students in a control group ($n = 61$) received no feedback. In the present study, the full sample was used for data analysis to increase statistical power. However, certain types of feedback affected students' strategy use and goal achievement (Theobald & Bellhäuser, 2022). Therefore, it was analysed whether the intervention had an effect on the hypothesized focal relations between perceived study time, study strategies, goal achievement and affect. First, it was tested whether experimental condition was a confounding factor in the analyses. For example, if the relation between study strategies and goal achievement is spurious and could be fully explained by experimental condition, then study strategies should no longer predict goal achievement once experimental condition is controlled for. However, the relations between study strategies, perceived study time, goal attainment and affect held when experimental condition was controlled for (see Tables 2 and 3). Second, it was tested whether experimental condition moderated the relations between study strategies, perceived study time and goal achievement. There was one significant interaction. The relation between self-monitoring strategies and goal achievement was stronger in the feedback conditions than in the control condition that received no feedback ($b = .002$, standardized $\beta = .04$, $SE = .001$, $p = .020$). This result must be taken into account when interpreting the results. Third, it was tested whether feedback moderated the proposed interactive effect of perceived study time and study strategies on goal achievement and affect (i.e. a three-way interaction between experimental condition, study strategies and perceived study time). There were

no three-way interactions. Taken together, these results suggest that the relations between perceived study time, study strategies, goal achievement and affect cannot be explained by experimental condition alone. Put differently, the reported effects may generalize independently of the provision of feedback (see also Discussion).

Descriptive statistics

Table 1 provides descriptive statistics and correlations between the focal variables. Perceived study time was related to goal achievement at the within-person level of analysis, but not at the between-person level. That is, average perceived study time was not related to average goal achievement. However, spending more time studying than the student usually does (within-person) was related to higher goal achievement. In addition, study strategies were related to goal achievement at both the within-person and between-person levels (except for planning). In addition, longer perceived study time was associated with more negative affect at the between-subject level, but with less negative affect at the within-subject level. On average, students who study longer report higher negative affect, but on days when students spent more time studying than usual (compared to their personal average), they report lower negative affect. In addition, study strategies (particularly low concentration and high procrastination) were related to higher negative affect at both the within-person and between-person levels. These results highlight the need to distinguish between the within-person and between-person levels, as the strength and direction of correlations may differ depending on the level of analysis.

Perceived study time and study strategies as predictors of daily goal achievement

First, longer study time on a given day was associated with higher goal achievement on that day ($\beta = .28$; see Table 2, Model 1). Second, better planning ($\beta = .04$), better monitoring ($\beta = .10$), better concentration ($\beta = .06$) and less procrastination ($\beta = -.31$) were associated with higher goal achievement (see Table 2, Model 2). Perceived study time remained a significant predictor of goal achievement after study strategies were included in the model. These results suggest that both study time and study strategies explain variance in daily goal achievement.

Finally, the central hypothesis that perceived study time and study strategies interactively predict daily goal achievement was tested (H1). Study strategies significantly moderated the relation between study time and goal achievement (see Table 2, Model 3 & Figure 2). For example, the relation between

TABLE 1 Means, standard deviations, within-person (grey-shaded) and between-person correlations.

Variable	<i>M</i>	<i>SD</i>	1	2	3	4	5	6	7
1. Goal achievement (%)	0.72	0.15		-.31**	.33**	.15**	.30**	.37**	-.48**
2. Negative affect	3.45	0.58	-.28**		-.04*	-.06**	-.19**	-.36**	.39**
3. Perceived study time (h)	3.69	1.55	.09	.16*		.17**	.30**	.13**	-.21**
4. Planning	4.45	0.74	.11	-.16*	.13*		.25**	.06**	-.11**
5. Monitoring	4.13	0.70	.23**	-.13	.12	.59**		.29**	-.28**
6. Concentration	3.71	0.71	.45**	-.40**	-.01	.12	.16*		-.69**
7. Procrastination	3.45	0.70	-.49**	.35**	-.16*	-.03	-.09	-.75**	

Note: Grey-shaded values indicate within-person correlations; unshaded coefficients indicate between-person correlations.

*Indicates $p < .05$.

**Indicates $p < .01$.

perceived study time and goal achievement was stronger when planning was low (1 SD below personal mean, $\beta = .22$, 95% CI [.14, .29], $p = .001$) compared to when planning was high (1 SD above the personal mean; $\beta = .33$, 95% CI [.24, .41], $p = .001$). That is, on days when students planned less, they had to invest more time to reach the same goal level compared to when they had planned their study day more carefully. A similar picture emerged for self-monitoring. Here, perceived study time was more strongly related to goal achievement when monitoring was low (vs. high; $\beta = .16$, 95% CI [.03, .28], $p = .012$). These results suggest that students may have compensated for lower planning and self-monitoring by investing more time. Regarding concentration, the relation between perceived study time and goal achievement was stronger when concentration was high (1 SD above personal mean, $\beta = .31$, 95% CI [.23, .38], $p = .001$) compared to when concentration was low (1 SD below the personal mean; $\beta = .23$, 95% CI [.15, .31], $p = .001$). In other words, the same amount of perceived study time spent with high concentration was associated with higher goal achievement than when it is spent with low concentration. Similarly, perceived study time was more strongly related to goal achievement when procrastination was high (vs. low; $\beta = -.35$, 95% CI [-.45, -.25], $p < .001$). Thus, when students procrastinated more on a given day, more perceived study time was required to achieve a similar goal level compared to days when students procrastinated less.

Notably, however, the significant association between perceived study time, planning and procrastination changed when perceived study time was particularly high. For example, when perceived study time exceeded 8 hours, higher levels of self-reported procrastination were no longer associated with goal achievement ($\beta = -.06$, 95% CI [-.21, .08], $p = .397$). For planning strategies, the association disappeared even earlier, at about 5 hours ($\beta = -.001$, 95% CI [-.07, .06], $p = .840$). A closer look at the data may provide an explanation for this finding. First, there were very few observations where perceived study time was greater than 8 hours ($k = 176$, 5% of the data), and these data points came from only a subset of participants ($n = 58$, 25%). Therefore, these data points may not be representative of the overall association across the sample. A more substantive explanation is that after a certain number of study hours, goals were almost always met, regardless of whether good strategies were used or not. Supporting this interpretation, the average goal achievement in this subset of the data was on average higher (about 82%) than in the rest of the distribution (about 73%). Thus, it was rare for those who planned their studies carefully to fail to meet their goals. However, with this high number of study hours, even those who did not plan their studies met their goals, so the advantage of planning the study day disappeared. Thus, at least for a subset of participants, studying for many hours may compensate for poorer study strategies.

Perceived study time and study strategies as predictors of daily negative affect

First, studying longer than usual (compared to the student's personal average) on a given day was associated with lower negative affect ($\beta = -.03$) (see Table 3, Model 1). Second, better monitoring ($\beta = -.07$), better concentration ($\beta = -.12$) and less procrastination ($\beta = .23$) compared to the student's personal average were associated with less negative affect on a given day (see Table 3, Model 2). Planning strategies did not predict affect. Perceived study time remained a significant predictor of goal achievement after study strategies were included in the model, but the direction of the relation reversed. After controlling for study strategies, studying more hours was associated with more negative affect. These results suggest that both perceived study time and study strategies explain variance in daily negative affect.

Finally, the hypothesis that perceived study time and study strategies interactively predict students' daily affect was tested (H2). Only concentration and procrastination moderated the relation between perceived study time and negative affect (see Table 3, Model 3, and Figure 3). When concentration was high (1 SD above the personal mean), students' perceived study time was not related to negative affect ($\beta = -.001$, 95% CI [-.19, .18], $p = .979$). However, when concentration was low (1 SD below the personal mean), negative affect increased as a function of perceived study time ($\beta = -.03$, 95% CI [-.05, -.01], $p = .039$). In other words, studying for many hours with low concentration was associated with higher negative affect. The results were somewhat different for procrastination. Here, for high levels

TABLE 2 Perceived study time and study strategies as predictors of daily goal achievement.

Coefficient	Model 1			Model 2			Model 3		
	Estimates	CI (95%)	p-value	Estimates	CI (95%)	p-value	Estimates	CI (95%)	p-value
Intercept	0.70	0.67–0.73	<.001	0.70	0.67–0.73	<.001	0.72	0.70–0.74	<.001
Exp. condition	0.01	–0.01 – 0.02	.337	0.01	–0.01 – 0.03	.279			
Study time	0.04	0.03–0.04	<.001	0.02	0.02–0.03	<.001	0.03	0.02–0.03	<.001
Planning				0.01	0.00–0.02	.008	0.01	0.00–0.02	.012
Monitoring				0.03	0.02–0.04	<.001	0.03	0.02–0.04	<.001
Concentration				0.01	0.00–0.02	.001	0.01	0.00–0.02	.002
Procrastination				–0.07	–0.07 – –0.06	<.001	–0.07	–0.07 – –0.06	<.001
Study time x Planning							–0.01	–0.01 – –0.00	.005
Study time x Monitoring							–0.00	–0.01 – –0.00	.038
Study time x Concentration							0.01	0.00–0.01	.001
Study time x Procrastination							0.01	0.01–0.02	<.001
Random effects									
σ^2	0.04			0.03			0.03		
τ_{00}	0.02 _{nr}			0.02 _{nr}			0.02 _{nr}		
ICC	0.28			0.35			0.34		
N	231 _{nr}			231 _{nr}			231 _{nr}		
Observations	3411			3411			3411		
Marginal R ² /Conditional R ²	.079/.333			.217/.487			.232/.496		

Note: p-values less than .05 indicate a statistically significant effect and are shown in bold.

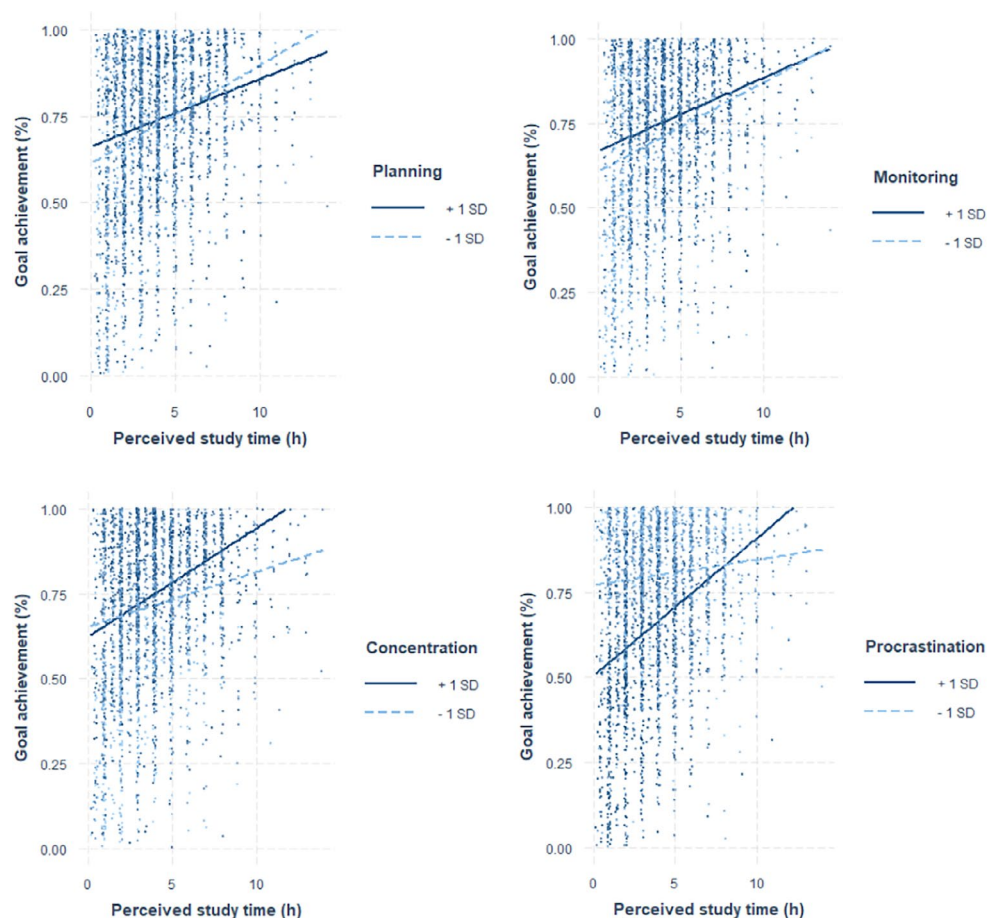


FIGURE 2 Relation between perceived study time (x-axis) and goal achievement (y-axis) for above-average study strategies (blue solid lines, 1 SD above mean) and below-average study strategies (blue dashed lines, 1 SD below mean).

of procrastination, perceived study time was not related to negative affect. In other words, 2 hours of studying and procrastination was associated with similar feelings as 8 hours of studying and procrastination ($\beta = .04$, 95% CI $[-.10, .17]$, $p = .590$). However, when procrastination was low, shorter perceived study time was associated with lower negative affect, and negative affect increased with the number of hours spent studying ($\beta = -.03$, 95% CI $[-.05, -.01]$, $p = .040$). Taken together, these results suggest that the negative affect was particularly high when students reported studying for many hours with low concentration; and it was particularly low when students reported studying for few hours and procrastinated less.

DISCUSSION

The present study tested the within-person interplay between study time and study strategies on daily goal achievement and affect at the within-person level. Results support the hypothesis that study time and study strategies interactively predict goal achievement. On days when students used better advance planning, more self-monitoring and less procrastination (compared to their personal mean), they showed higher goal achievement in less time. Thus, studying longer was associated with better goal achievement, especially when that time was spent using good study strategies. In addition, concentrated studying even strengthened the link between perceived study time and goal

TABLE 3 Perceived study time and study strategies as predictors of daily affect.

Coefficient	Model 1			Model 2			Model 3		
	Estimates	CI (95%)	p-value	Estimates	CI (95%)	p-value	Estimates	CI (95%)	p-value
Intercept	3.52	3.40–3.65	<.001	3.53	3.41–3.66	<.001	3.52	3.40–3.65	<.001
Exp. condition	–0.06	–0.12 – 0.01	.102	–0.06	–0.12 – 0.01	.085	–0.06	–0.12 – 0.01	.082
Study time	–0.01	–0.03 – –0.00	.030	0.03	0.01–0.04	<.001	0.03	0.01–0.04	<.001
Planning				–0.00	–0.03 – 0.03	.779	–0.00	–0.03 – 0.03	.779
Monitoring				–0.08	–0.11 – –0.04	<.001	–0.08	–0.11 – –0.04	<.001
Concentration				–0.10	–0.13 – –0.07	<.001	–0.10	–0.13 – –0.07	<.001
Procrastination				0.19	0.16–0.22	<.001	0.19	0.16–0.22	<.001
Study time x Planning							–0.01	–0.03 – 0.01	.251
Study time x Monitoring							0.00	–0.01 – 0.02	.789
Study time x Concentration							–0.02	–0.04 – –0.00	.014
Study time x Procrastination							–0.03	–0.04 – –0.01	<.001
Random effects									
σ^2	0.55			0.45			0.45		
τ_{00}	0.28 _{nr}			0.29 _{nr}			0.28 _{nr}		
ICC	0.34			0.39			0.39		
N	231 _{nr}			231 _{nr}			231 _{nr}		
Observations	3411			3411			3411		
Marginal R ² /Conditional R ²	.006/.342			.124/.465			.127/.465		

Note: p-values less than .05 indicate a statistically significant effect and are shown in bold.

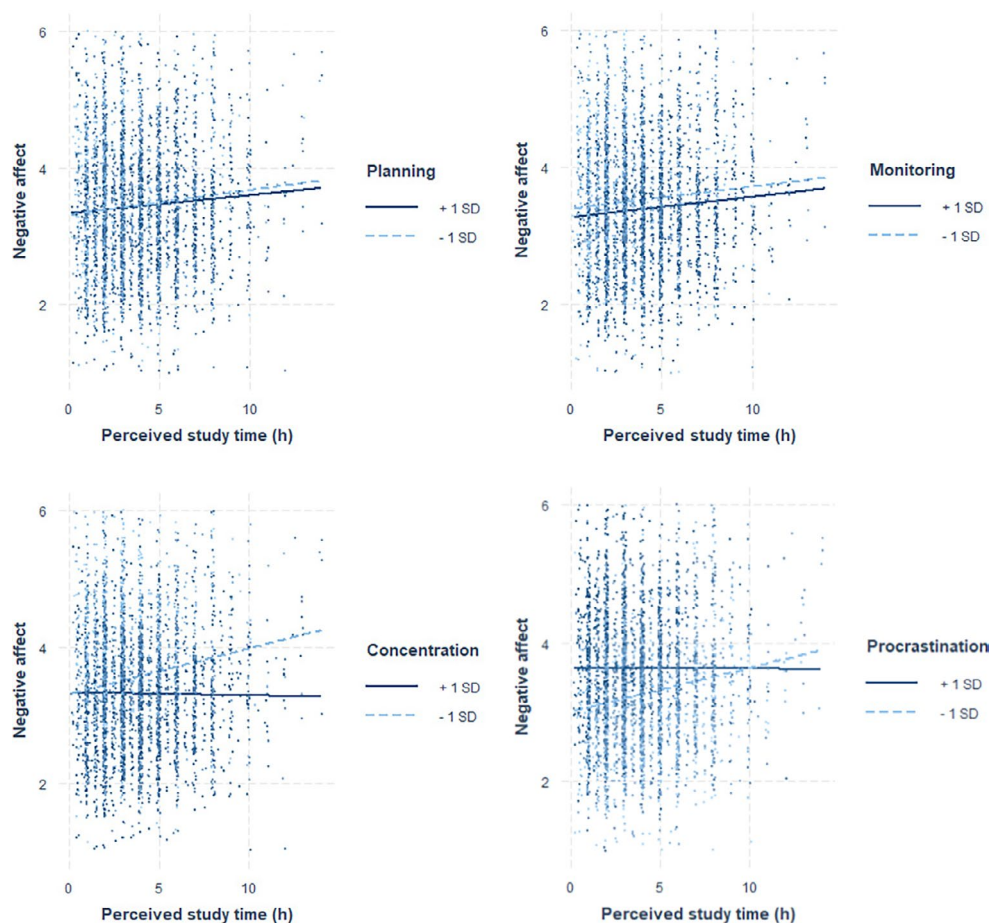


FIGURE 3 Relation between perceived study time (x-axis) and negative affect (y-axis) for above-average study strategies (blue solid lines, 1 SD above mean) and below-average study strategies (blue dashed lines, 1 SD below mean).

achievement. Thus, when concentration is high, studying longer may be associated with increased goal achievement. In terms of affect, the results showed that, controlling for study strategies, studying longer was associated with more negative affect. Negative affect was particularly high when students reported studying more than usual and with less concentration than usual and it was particularly low when students reported studying for fewer hours than usual and procrastinating less than usual. In summary, the results provide initial evidence for the interactive interplay between study time and study strategies on daily goal achievement. The results suggest that better study strategies may promote goal achievement, but may reduce negative affect and overall study load in terms of hours spent studying.

The findings support models of self-regulated learning that emphasize the importance of studying strategically. Consistent with previous evidence, students who studied longer and more strategically (compared to their own average) reported better learning outcomes (Credé & Kuncel, 2008; Richardson et al., 2012). Importantly, however, study time predicted better learning outcomes, especially when it was used effectively (Schmitz & Wiese, 2006). These findings support an interactive relation between study time and study strategies on learning outcomes. Despite its plausibility, evidence for this interactive relation has been scarce (see Doumen et al., 2014 for an exception). In particular, evidence for the within-person interaction between study time and study strategies in predicting daily fluctuations

in goal achievement has been lacking. Students' study time, study strategies and goal achievement have been shown to vary from one study session to the next (Liborius et al., 2019; Theobald et al., 2023). Furthermore, in the present study, the strength and direction of correlations often differed at the between-subject and within-subject levels of analysis, highlighting the clear need to distinguish between within-person and between-person relations. Thus, future research could compare the interplay between study time and study strategies at the within-person and between-person levels. In summary, the present study fills an important gap in the literature by demonstrating the interactive relation of study time and study strategies on goal achievement at the daily within-person level.

The mechanisms by which planning, self-monitoring, concentration and procrastination contribute to more efficient use of time may differ. For example, planning may help learners to set concrete goals and priorities for studying, while self-monitoring may facilitate the comparison between these plans and current progress during studying (Wolters & Brady, 2020). Thus, proactive strategies such as planning and self-monitoring may help students to achieve the same learning outcomes in less time. Improved concentration may support the avoidance of distractions during learning (Plant et al., 2005). In the present study, on days when students were better able to concentrate, increased study time benefited goal achievement. Here, studying longer may have led students to enter a 'flow' state, which further enhanced goal achievement. Regarding procrastination, students who do not procrastinate begin a study task earlier (Steel, 2007). In contrast, students who procrastinate more need to spend more time studying to compensate for the procrastination. In summary, planning, self-monitoring, concentration and avoidance of procrastination may have contributed to more efficient use of time by supporting various self-regulatory processes. However, experimental studies are needed to examine the role of specific study strategies and the underlying mechanisms by which they support more efficient time use.

When students do not study strategically, they may have to study longer to reach a similar goal level. The results showed that, at least for some students who studied many hours on a given day (>8 h), study strategies did not further increase goal achievement. Thus, these students may have compensated for poorer study strategies by studying longer. The results show that studying longer is not necessarily bad, as it can also improve goal achievement. However, there are drawbacks to studying longer. A high number of study hours inevitably reduces the time available for other relaxing activities, hobbies or spending time with friends. As shown in the present and previous studies, studying many hours per day can increase negative emotions and stress (Lutz-Kopp et al., 2019; Theobald & Bellhäuser, 2021). This association between high study time and negative affect may be exacerbated if students spent this time using ineffective study strategies. When students studied for many hours with low concentration or when they procrastinated more than usual, their negative affect was particularly high. Therefore, these findings underscore the need to teach students good study strategies not only to promote goal achievement in less time but also to counteract negative affect and stress.

Although the present study clearly supports that studying strategically is more beneficial than studying longer, these results must be considered with limitations. First, the sample consisted only of university students. University students often study independently, making them an ideal sample to test the interaction between study time and study strategies. Future studies should also test the interactive effect of study time and study strategies in other samples, such as school children. Second, the present study included an experimental manipulation. Some students received feedback on their daily study strategies and motivation. It was therefore tested whether the experimental manipulation affected the correlations between our focal variables of interest. The link between self-monitoring and goal achievement was slightly stronger in the feedback conditions compared to the control condition. These results suggest that the proposed relations between study time, study strategies and goal achievement mostly hold regardless of the experimental manipulation. Nevertheless, future studies should replicate the findings in a sample that was not part of an experimental setting. Another limitation is the correlational and self-report nature of the data. Therefore, experimental studies are needed to test causal relations. In addition, future studies should assess study time and goal achievement using more fine-grained and objective measures. For example, students' perceived study time may differ from their actual study time. Moreover, students had to rate their average goal achievement across potentially multiple goals, which

calls for a more nuanced assessment of students' goal achievement in future research. Finally, given that the within-person reliability for the planning and self-monitoring measure was somewhat low, future studies should develop and test self-report scales that more reliably assess intra-individual differences in planning and self-monitoring skills.

In conclusion, this study examined the often proposed but rarely tested hypothesis that studying strategically can compensate for shorter study time. The results support this hypothesis by showing that on days when students studied more strategically than they usually do, they took less time to achieve their goals. Effective use of time may also reduce the negative affective consequences associated with a large number of (unfocused) study hours. These findings underscore the clear need to teach students effective study strategies. However, many university students struggle to manage their study time effectively, as evidenced by the high rates of procrastination (Klingsieck, 2013; Steel, 2007). Therefore, training programs could be offered to promote students' use of strategies. There is meta-analytic evidence that self-regulated learning strategies can improve students' study strategies such as planning and monitoring (Theobald, 2021). Therefore, teaching students effective study strategies has the potential to improve their goal achievement while reducing study load and negative affect.

AUTHOR CONTRIBUTIONS

Maria Theobald: Conceptualization; investigation; funding acquisition; writing – original draft; methodology; visualization; writing – review and editing; formal analysis; project administration; data curation; resources.

ACKNOWLEDGEMENTS

This research was supported by a university research grant from the Johannes Gutenberg-University Mainz, Germany (Center for School, Education, and Higher Education Research).

CONFLICT OF INTEREST STATEMENT

The author has no conflict of interest to declare.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in Open Science Framework at <https://osf.io/r9n4y/>.

TRANSPARENCY AND OPENNESS

The data, the data analysis script and a codebook are available via the Open Science Framework (Link to project: <https://osf.io/r9n4y/>).

ORCID

Maria Theobald  <https://orcid.org/0000-0003-0766-1680>

REFERENCES

- Boekaerts, M. (1999). Self-regulated learning: Where we are today. *International Journal of Educational Research*, 31, 445–457. [https://doi.org/10.1016/S0883-0355\(99\)00014-2](https://doi.org/10.1016/S0883-0355(99)00014-2)
- Broadbent, J., & Poon, W. L. (2015). Self-regulated learning strategies & academic achievement in online higher education learning environments: A systematic review. *The Internet and Higher Education*, 27, 1–13. <https://doi.org/10.1016/j.iheduc.2015.04.007>
- Credé, M., & Kuncel, N. R. (2008). Study habits, skills, and attitudes: The third pillar supporting collegiate academic performance. *Perspectives on Psychological Science*, 3(6), 425–453. <https://doi.org/10.1111/j.1745-6924.2008.00089.x>
- Dent, A. L., & Koenka, A. C. (2016). The relation between self-regulated learning and academic achievement across childhood and adolescence: A meta-analysis. *Educational Psychology Review*, 28(3), 425–474. <https://doi.org/10.1007/s10648-015-9320-8>
- Doumen, S., Broeckmans, J., & Masui, C. (2014). The role of self-study time in freshmen's achievement. *Educational Psychology*, 34(3), 385–402. <https://doi.org/10.1080/01443410.2013.785063>
- Galloway, M., Conner, J., & Pope, D. (2013). Nonacademic effects of homework in privileged, high-performing high schools. *The Journal of Experimental Education*, 81(4), 490–510. <https://doi.org/10.1080/00220973.2012.745469>

- Glöckner-Rist, A., Engberding, M., Höcker, A., & Rist, F. (2014). Prokrastinationsfragebogen für Studierende (PFS) [procrastination questionnaire for students]. *Zusammenstellung Sozialwissenschaftlicher Items Und Skalen/Compilation of Social Science Items And Scales*. <https://doi.org/10.6102/zis140>
- Häfner, A., Stock, A., Pinneker, L., & Ströhle, S. (2014). Stress prevention through a time management training intervention: An experimental study. *Educational Psychology*, 34(3), 403–416. <https://doi.org/10.1080/01443410.2013.785065>
- Klingsieck, K. B. (2013). Procrastination when good things don't come to those who wait. *European Psychologist*, 18(1), 24–34. <https://doi.org/10.1027/1016-9040/a000138>
- Klug, J., Ogrin, S., Keller, S., Ihringer, A., & Schmitz, B. (2011). A plea for self-regulated learning as a process: Modelling, measuring and intervening. *Psychological Test and Assessment Modeling*, 53(1), 51–72.
- Landrum, R. E., Turrissi, R., & Brandel, J. M. (2006). College Students' study time: Course level, time of semester, and grade earned. *Psychological Reports*, 98(3), 675–682. <https://doi.org/10.2466/pr0.98.3.675-682>
- Leiner, D. J. (2019). SoSci Survey [Software]. <https://www.sosicisurvey.de>
- Liborius, P., Bellhäuser, H., & Schmitz, B. (2019). What makes a good study day? An intraindividual study on university students' time investment by means of time-series analyses. *Learning and Instruction*, 60, 310–321. <https://doi.org/10.1016/j.learninstruc.2017.10.006>
- Lutz-Kopp, C., Meinhardt-Injac, B., & Luka-Krausgrill, U. (2019). Psychische Belastung Studierender. *Prävention Und Gesundheitsförderung*, 14(3), 256–263. <https://doi.org/10.1007/s11553-018-0691-9>
- Macan, T. H., Shahani, C., Dipboye, R. L., & Phillips, A. P. (1990). College students' time management: Correlations with academic performance and stress. *Journal of Educational Psychology*, 82(4), 760–768. <https://doi.org/10.1037/0022-0663.82.4.760>
- Molenaar, P. C. M. (2004). A manifesto on psychology as idiographic science: Bringing the person Back into scientific psychology, this time forever. *Measurement: Interdisciplinary Research and Perspectives*, 2(4), 201–218. https://doi.org/10.1207/s15366359mea0204_1
- Panadero, E. (2017). A review of self-regulated learning: Six models and four directions for research. *Frontiers in Psychology*, 8, 1–28. <https://doi.org/10.3389/fpsyg.2017.00422>
- Pintrich, P. R. (1999). The role of motivation in promoting and sustaining self-regulated learning. *International Journal of Educational Research*, 31, 459–470. [https://doi.org/10.1016/S0883-0355\(99\)00015-4](https://doi.org/10.1016/S0883-0355(99)00015-4)
- Plant, E. A., Ericsson, K. A., Hill, L., & Asberg, K. (2005). Why study time does not predict grade point average across college students: Implications of deliberate practice for academic performance. *Contemporary Educational Psychology*, 30(1), 96–116. <https://doi.org/10.1016/j.cedpsych.2004.06.001>
- Richardson, M., Abraham, C., & Bond, R. (2012). Psychological correlates of university students' academic performance: A systematic review and meta-analysis. *Psychological Bulletin*, 138(2), 353–387. <https://doi.org/10.1037/a0026838>
- Schallberger, U. (2005). Kurzskalen zur Erfassung der Positiven Aktivierung, Negativen Aktivierung und Valenz in Experience Sampling Studien (PANAVA-KS). In *Forschungsberichte aus dem Projekt: "Qualität des Erlebens in Arbeit und Freizeit"* (Vol. 6, pp. 1–87). Psychologisches Institut.
- Schmitz, B., & Wiese, B. S. (2006). New perspectives for the evaluation of training sessions in self-regulated learning: Time-series analyses of diary data. *Contemporary Educational Psychology*, 31(1), 64–96. <https://doi.org/10.1016/j.cedpsych.2005.02.002>
- Sirois, F. M. (2023). Procrastination and stress: A conceptual review of why context matters. *International Journal of Environmental Research and Public Health*, 20(6), 5031. <https://doi.org/10.3390/ijerph20065031>
- Steel, P. (2007). The nature of procrastination: A meta-analytic and theoretical review of quintessential self-regulatory failure. *Psychological Bulletin*, 133(1), 65–94. <https://doi.org/10.1037/0033-2909.133.1.65>
- Theobald, M. (2021). Self-regulated learning training programs enhance university students' academic performance, self-regulated learning strategies, and motivation: A meta-analysis. *Contemporary Educational Psychology*, 66, 101976. <https://doi.org/10.1016/j.cedpsych.2021.101976>
- Theobald, M., Bäumle, L., Bellhäuser, H., Breitwieser, J., Mattes, B., Brod, G., Daumiller, M., Dresel, M., Liborius, P., Nückles, M., Bellh, H., Theobald, M., Lisa, B., Brod, G., Daumiller, M., Dresel, M., Liborius, P., & Nückles, M. (2023). A multi-study examination of intra-individual feedback loops between competence and value beliefs, procrastination, and goal achievement. *Contemporary Educational Psychology*, 74, 102208. <https://doi.org/10.1016/j.cedpsych.2023.102208>
- Theobald, M., & Bellhäuser, H. (2021). Mind the gap! Unmet time schedules predict university students' negative affect during the examination phase. *Zeitschrift Für Entwicklungspsychologie Und Pädagogische Psychologie (German Journal of Developmental and Educational Psychology)*, 51(4), 190–203. <https://doi.org/10.1026/0049-8637/a000218>
- Theobald, M., & Bellhäuser, H. (2022). How am I going and where to next? Elaborated online feedback improves university students' self-regulated learning and performance. *The Internet and Higher Education*, 55, 100872. <https://doi.org/10.1016/j.iheduc.2022.100872>
- Theobald, M., Bellhäuser, H., & Imhof, M. (2018). Identifying individual differences using log-file analysis: Distributed learning as mediator between conscientiousness and exam grades. *Learning and Individual Differences*, 65, 112–122. <https://doi.org/10.1016/j.lindif.2018.05.019>
- Theobald, M., Breitwieser, J., Murayama, K., & Brod, G. (2021). Achievement emotions mediate the link between goal failure and goal revision: Evidence from digital learning environments. *Computers in Human Behavior*, 119, 106726. <https://doi.org/10.1016/j.chb.2021.106726>

- Tice, D. M., & Baumeister, R. F. (1997). Longitudinal study of procrastination, performance, stress, and health: The costs and benefits of dawdling. *Psychological Science*, 8(6), 454–458. <https://doi.org/10.1111/j.1467-9280.1997.tb00460.x>
- Veenman, M. V. J., van Hout-Wolters, B. H. A. M., & Afflerbach, P. (2006). Metacognition and learning: Conceptual and methodological considerations. *Metacognition and Learning*, 1(1), 3–14. <https://doi.org/10.1007/s11409-006-6893-0>
- Wild, K. P., & Schiefele, U. (1994). Lernstrategien im Studium: Ergebnisse zur Faktorenstruktur und Reliabilität eines neuen Fragebogens [learning strategies for students: Factor Structur and reliability of a new questionnaire]. *Zeitschrift für Differentielle und Diagnostische Psychologie/German Journal For Differential and Diagnostic Psychology*, 15, 185–200.
- Wolters, C. A., & Brady, A. C. (2020). College Students' time management: A self-regulated learning perspective. *Educational Psychology Review*, 33, 1319–1351. <https://doi.org/10.1007/s10648-020-09519-z>

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Theobald, M. (2025). Study longer or study effectively? Better study strategies can compensate for less study time and predict goal achievement and lower negative affect. *British Journal of Educational Psychology*, 95, 405–420. <https://doi.org/10.1111/bjep.12725>